

§15.6 Surface Area

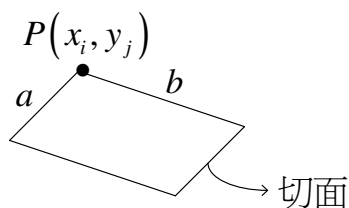
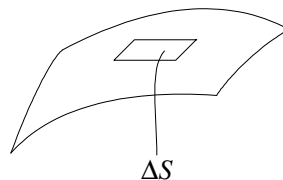
Homework : 1,5,11,13,21,23

單變數函數

弧長 → 用線段逼近

多變數函數

Surface Area → 切面面積逼近



過 P_{ij} 的切面來逼近.

過 P_{ij} 點的曲面.

$$a = \Delta x i + o j + f_x \Delta x k$$

$$b = o i + \Delta y j + f_y \Delta y k$$

曲面面積 = $\Delta S \approx$ 切面面積

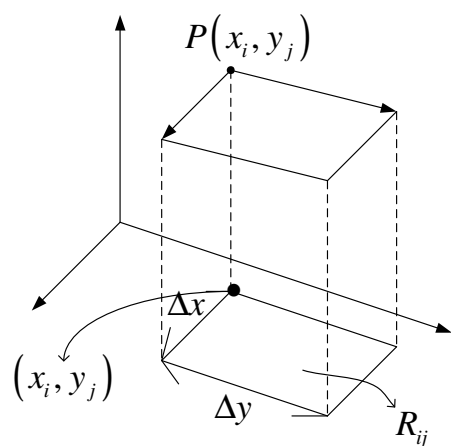
$$\Rightarrow \text{切面面積} = |a \times b|$$

$$= \sqrt{1 + f_x^2 + f_y^2} (\Delta x \Delta y)$$

$$= \sqrt{1 + f_x^2 + f_y^2} (\Delta A)$$

⇒ 作無窮切割後

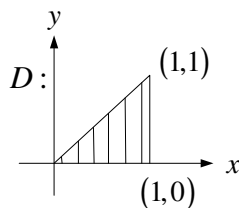
$$dS = \sqrt{1 + f_x^2 + f_y^2} dA$$



公式：地基為 D ，屋頂為 $z = f(x, y)$

$$\Rightarrow \text{其屋頂面積} = S = \iint_D dS = \iint_D \sqrt{1 + f_x^2 + f_y^2} dA$$

Example 1 : $z = x^2 + 2y$.



Solution :

$$\Rightarrow S = \int_0^1 \int_x^1 \sqrt{1 + (2x)^2 + 2^2} dy dx = \frac{1}{12} (27 - 5\sqrt{5})$$

Example 2 : Find surface area of $z = x^2 + y^2$ under $z = 9$.

Solution :

$$\begin{aligned} S &= \iint_{x^2+y^2 \leq 9} \sqrt{1+4x^2+4y^2} \, dA \\ &= \int_0^{2\pi} \int_0^3 \sqrt{1+4r^2} \, r \, dr \, d\theta \\ &= \frac{\pi}{6} (37\sqrt{37} - 1) \end{aligned}$$

Example 3 : Find surface area of $z = y^2 - x^2$ between $x^2 + y^2 = 1$ and $x^2 + y^2 = 4$.

Solution :

$$S = \iint_{1 \leq x^2+y^2 \leq 4} \sqrt{1+4y^2+4x^2} \, dA = \text{極座標轉換}$$

Example 4 : Find the area of the finite part of the paraboloid $y = x^2 + z^2$ cut off by the plane $y = 25$.

Solution :

$$\begin{aligned} \text{地基} \quad & \begin{cases} x^2 + z^2 = y \\ y = 25 \end{cases} \Rightarrow x^2 + z^2 = 5^2 \quad (y=0) \\ \text{屋頂} \quad & y = x^2 + z^2 \end{aligned}$$

$$\begin{aligned} S &= \iint_{x^2+z^2 \leq 5^2} \sqrt{1+4x^2+4z^2} \, dA \\ &= \int_0^{2\pi} \int_0^5 r \sqrt{1+4r^2} \, dr \, d\theta \\ &= \frac{\pi}{6} (101\sqrt{101} - 1) \end{aligned}$$